
Algorithm 1 ARCH-Major: Adaptive Recursive Clustering Algorithm

Require: Major name list $items$, pre-trained model $model$, parameters

τ, θ, M, D

Ensure: Clustering result dictionary $clusters$

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1: Function ARCH-MAJOR( $items, model, \tau, \theta, M, D$ )
2:    $clusters \leftarrow \emptyset$ 
3:    $initial\_clusters \leftarrow \text{RECURSIVE-CLUSTER}(items, model, \tau, \theta, M, D, 0)$ 
4:    $next\_id \leftarrow 0$ 
5:   For  $(label, cluster) \in initial\_clusters$  do
6:     If  $label = -1 \vee |cluster| > M$  then
7:        $refined \leftarrow \text{REFINE-LARGE-CLUSTER}(cluster, model, \tau +$ 
8:          $0.1, \theta + 0.05, M, D)$ 
9:       For  $(sub\_label, sub\_cluster) \in refined$  do
10:        If  $sub\_label \neq -1$  then
11:           $clusters[next\_id] \leftarrow sub\_cluster$ 
12:           $next\_id \leftarrow next\_id + 1$ 
13:        Else
14:           $clusters[-1] \leftarrow clusters[-1] \cup sub\_cluster$ 
15:        End If
16:      End For
17:    Else
18:       $clusters[next\_id] \leftarrow cluster$ 
19:       $next\_id \leftarrow next\_id + 1$ 
20:    End If
21:  End For
22:  Return  $clusters$ 
23: End Function

23: Function RECURSIVE-CLUSTER( $items, model, \tau, \theta, M, D, depth$ )
24:  If  $|items| \leq M \vee depth \geq D$  then
25:    Return BASE-CLUSTER( $items$ )  $\triangleright$  Base clustering method
26:  End If
27:   $embeddings \leftarrow model.encode(items)$ 
28:   $S \leftarrow \text{COSINE-SIMILARITY}(embeddings)$ 
29:   $labels \leftarrow \text{DBSCAN}(embeddings, \epsilon, min\_samples)$ 
30:   $clusters \leftarrow \emptyset$  1
31:  For  $label \in \text{UNIQUE}(labels)$  do
32:     $cluster\_items \leftarrow \{items[i] \mid labels[i] = label\}$ 
33:    If  $label = -1 \vee |cluster\_items| > M$  then
34:       $sub\_clusters \leftarrow \text{RECURSIVE-CLUSTER}(cluster\_items, model, \tau, \theta, M, D, depth +$ 
35:         $1)$ 
36:       $clusters \leftarrow clusters \cup sub\_clusters$ 
37:    Else
38:       $components \leftarrow \text{CHECK-CONNECTIVITY}(S, \tau)$ 
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